More or Less: How Information Richness Affects Our Decision Consistency

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B2B e-commerce platforms increasingly mediate supplier selection by displaying extensive attribute information. However, excessive information may distort customer decision-making and reduce alignment between stated preferences and actual choices. Across three controlled experiments, we examine how information volume and sequencing influence decision consistency. We find that customers demonstrate greater consistency between stated and revealed preferences when supplier attributes are fewer. Moreover, introducing users first to simplified information before presenting complex profiles significantly improves consistency. These effects are robust across designs, populations, and measurement approaches. Our results suggest that progressive disclosure of supplier attributes—a simple and scalable interface design—can enhance decision stability and mitigate distortions from information overload on B2B platforms.

Key words: information volume, decision consistency, revealed preference, stated preference, e-commerce platform

1. Introduction

Emerging Business-to-Business (B2B) e-commerce platforms allow retailers to enjoy the convenience of locating manufacturing suppliers online. In 2020, the global B2B e-commerce market size was valued at USD 6.64 trillion and is expected to expand at an annual growth rate of 18.7% from 2021 to 2028.[1] On B2B e-commerce platforms, manufacturers can sell not only finished goods but also their available production capacity; retail customers now have an extensive volume of information available at their fingertips that enables them to evaluate thousands of suppliers regardless of their geographical locations or prior knowledge of the supply base.

Most customers in B2B e-commerce are composed of millennials, born between 1981 and 1996 (ages 25 to 40 in 2021), who contribute to the majority of the workforce. Since this population

[1] Grand View Research Market Analysis Report: Business-to-Business E-commerce Market Size, Share and Trends Analysis Report By Deployment Model (Intermediary-oriented, Supplier-oriented), By Application, By Region, And Segment Forecasts, 2021 - 2028.

is quite familiar with the Business-to-Customer (B2C) online shopping experience, they often expect and want the same experience from B2B websites. They want the suppliers to be clearly characterized with standardized information about their attributes such as price, quality, and shipping fee, just as how B2C websites list such standardized information about their products. They also want such information to be cleanly organized and easily legible.

However, the user interface design of a B2B website cannot readily be directly copied from that of a B2C website, due to the type of users on the platforms and the extended nature of the transactions. First, the attributes needed to characterize a supplier are more complex compared to the limited attributes needed to specify a B2C-listed product. For example, Alibaba's Tao Factory is now the largest B2B platform in China and is still growing quickly. Sellers on the platform are factories that sell their available production capacity online. For each listing, there are as many as 27 different attributes, including unit sourcing price, quality of production, speed of response, production and shipping, as well as various service guarantees for ensuring material, speed of production, and so on. It requires enormous information processing efforts for retailers to sort through such an extensive list of attributes to locate the desired suppliers. Second, business customers' preferences for various attributes can be quite different from individual customers' preferences. For example, a free return policy could be one of the most critical attributes for an individual customer who considers the money-back guarantee to afford them zero risk of loss when placing the order. However, for business customers who purchase a large volume of products, a free return policy would still not prevent loss associated with the opportunity cost of capital, missing valuable selling windows, and reputation damages due to backlogs. In addition, given the large amount of attributes, customers can form their own purchasing criteria by sorting the attributes' importance. Retail customers may either weigh all available attributes or rely selectively on a few salient ones, highlighting the complexity of preference construction under cognitive load. The possible composition of selection criteria is more variable compared to that of individual customers.

Hence, how to compose and arrange the user interface design is a critical question to resolve for B2B e-commerce. Addressing this question has direct implications for platform efficiency, supplier-buyer matching quality, and the operational performance of digital procurement systems. After a user interface upgrade, the Alibaba Tao Factory has changed the number of criteria displayed on the platform. Comparing how customers interact with the user interface from the Alibaba Tao Factory before and after the upgrade, we notice that the amount of information displayed likely changes customers' decision criteria (see Figure 1). When the platform downsized the quantity of the posted attributes by half, using the gradient-boosted decision trees model, customers' emphasis on attributes changed in response. When faced with all of the 27 supplier attributes at once, the customers tended to consider only a few attributes as important for locating the suppliers

Figure 1 Change in Decision Criteria under Two Information Designs: Left Panel (More Posted Attributes), Right Panel (Fewer Posted Attributes)



(right panel of Figure 1). However, when faced with only 14 attributes, the customers utilized all available attributes to make decisions (left panel of Figure 1). Unfortunately, because suppliers can endogenously decide whether to adopt the new user interface, we cannot directly rely on the Alibaba interface change to causally evaluate the impact of information volume on supplier selection criteria. This limitation motivates our experimental approach, which isolates the causal impact of information volume and sequence while controlling for contextual confounds present in real-world settings.

Motivated by this observation, in this research, we primarily focus on the role of information volume on decision consistency. Specifically, our research investigates three questions. The first is how retailers allocate weights on supplier attributes based on the volume of information presented. The second and more important question is about how retailers' allocated weights (i.e., their actual preference for certain supplier attributes over others when making the choice) may differ from their stated preference due to the volume of information presented. Investigation of the first question reveals how the volume of information influences decision criteria used by retailers. The second question digs deeper into the decision criteria by revealing unexpected inconsistency between the criteria that retailers act upon (their actual preferences) and the criteria that retailers think they act upon (their stated preferences).

We choose to adopt the lab experiment method to identify the role of information volume in decision inconsistency between stated and actual preference. The lab experiment method facilitates our study in two critical ways. First, it allows us to record decision-makers' stated preferences, which is hardly retrievable in real-world supplier selection decisions. Based on the ranking provided by the experiment participants, we can identify the list of attributes they consider most important and the list of attributes they consider least important. In addition, the lab experiment method enables us to control for multiple confounding factors in selection criteria. In the experiment, we provided all participants with the same supplier selection setting and thus ruled out the potential impact

of heterogeneous customer requests that complicate real-life observations. Compared to real B2B selections, where customers seek to fulfill their business needs while facing information overload from 1) the sheer amount of available supplier attributes and 2) the extensive number of candidate suppliers, we distinguish the two types of information processing by holding constant the number of candidate suppliers and focusing on the variation in the number of supplier attributes. Our study reveals actionable insights into how information design influences consistency in decision-making. First, we confirm that information volume affects customers' selection criteria in the B2B setting. The inconsistency between participants' stated and revealed preferences varies with the number of supplier attributes provided. Participants demonstrate greater consistency when faced with a short list of attributes than a long list, suggesting that higher information volumes exacerbate cognitive challenges in aligning stated preferences with actual decision-making. Our study then specifies an effective mechanism design for B2B user interfaces to correct such inconsistency in information usage. We observe that retailers who are first exposed to a shorter list of attributes can carry their consistent supplier selection decisions into more complex decision making when subsequently viewing a longer list of attributes. Hence, B2B platforms can progressively display information to enable decision-makers to follow through with their stated selection criteria more consistently.

Our research contributes to academic studies and managerial practices in multiple ways. Our research makes three contributions to the literature on operations management and decision science. First, we introduce and quantify a novel operational construct: *preference inconsistency at the attribute level* under varying information conditions. Second, we establish a causal link between information volume and decision inconsistency, revealing that increased attribute load undermines the alignment between stated preferences and actual selection behaviour. Third, we demonstrate that progressive information sequencing—starting from fewer attributes—offers a viable interface design strategy to mitigate inconsistency. These findings contribute to supplier selection research, information system design, and behavioural operations, with practical implications for digital procurement platforms seeking to improve customer decision quality. Practically, we not only identify the role that information volume plays in preference consistency, but we also identify an effective mechanism design to reduce decision inconsistency. Our results show that offering decision-makers progressive amounts of information can help them act upon their stated preferences more consistently. From a practical perspective, our research offers insights regarding how to more effectively characterize products' attributes on B2B platforms.

The rest of the paper is organized in the following way. In Section 2, we review the extant literature on information volume and preference inconsistency and raise our hypothesis. In Section 3, we describe our lab experiment setup. Then we discuss our empirical methods for analyzing the data in Section 4 and our results in Section 5. Finally, we discuss the managerial implications for the practical application of our findings in Section 6.

2. Literature Review

Whether humans are rational creatures has been investigated by researchers for a long time throughout history. Literature on human information process theory can be dated back to rational choice theory in economic studies. Rational choice theory views humans as "Homo economicus," or economic man. It assumes that humans are consistently rational and pursue utility optimization only in their interest (Zak and Jensen 2010).

However, this view of human' economic rationality has been challenged by a series of studies in both economics and psychology. Herbert Simon was the first to introduce the principle of "bounded rationality" as a call to replace the classical view of human rationality. Bounded rationality states that people are not cognitively unlimited agents. Instead, the human decision-making process is constrained by cognitive ability and by the limits of circumstances (Simon 1956, Simon et al. 1989, Simon 1990). Psychologists have made many attempts to measure the amount of information people can hold in short-term memory. Miller (1956) first quantified the working memory capacity for processing information as limited to seven chunks of elements. Several other studies have revised the measurement; for example, Broadbent (1975) proposed a working memory capacity of three to four elements.

2.1. Decision making affected by information volume

Since the 1950s, it has been widely accepted that human cognitive capacity is limited. Daniel Kahneman and Amos Tversky have suggested that a large portion of human decision making is not rational. Kahneman and Tversky showed that people rely on simple mental shortcuts called heuristics to evaluate possibilities and frequencies (Kahneman et al. 1982). As the theory of limited cognitive processing capability developed, many empirical studies have investigated the impact of information volume on decision-making. Existing research on information overload has primarily operationalised information volume as the number of available alternatives rather than the number of attributes per option. It explored the impact of a large set of choices on decision quality, motivation to make decisions, and emotion during decision-making. For example, existing research has shown the choice overload effect, where as the number of alternatives increases, the probability of choosing the best one decreases (Lurie 2004). In addition to affecting decision quality, existing research also indicates that a large set of choices influences decision-makers' motivation to make a choice and their emotional state during the decision-making process. Customers tend to show satisfaction with extensive choices but are less motivated to purchase when asked to make a purchase. Iyengar and Lepper (2000) showed that customers were more attracted to a tasting booth when there were 24 different jams than when there were six choices. However, customers in the extensive-choice condition were much less likely to buy a jar of jam after tasting it than those with limited

choices. Long et al. (2024) also observed a decrease in consumers' search and purchasing probabilities as the number of recommended products increased on an online retail platform. In addition to influencing purchasing behavior, people are more likely to have a higher level of negative emotions such as regret and disappointment after making decisions in a choice-extensive context (Schwartz 2000). Studies have also shown that when information volume is high, people tend to selectively attend to some attributes, rely on heuristics, or perform "social learning," simply choosing what everyone else is choosing (Tversky 1972, Efferson et al. 2008).

Some other papers defined information as the multiplier of the number of choices and the number of attributes of each choice and yielded similar results (Jacoby et al. 1974, Malhotra 1982). For example, more information leads to a higher mental load, which will negatively impact choice quality Jacoby et al. (1974). For example, Malhotra (1982) showed that when the number of attributes exceeds 15, respondents start to exhibit the dysfunctional effect of information volume. More recently, Lee and Lee (2004) replicated these results, demonstrating that when customers are faced with 18 attributes instead of 9 attributes, they make significantly poorer decisions in an online setting. Our study extends this literature in two key ways by (1) isolating attribute-level information load while holding the number of alternatives constant, and (2) shifting focus from decision quality defined by correctness to decision consistency defined by preference stability. First, we define information overload as the number of attributes per option. Consequently, we manipulate only the number of attributes per option while maintaining a constant number of options. Second, and more importantly, we aim to explore how the volume of information affects decision consistency, namely, the discrepancy between stated and actual preferences. Unlike previous studies that define decision quality as the likelihood of selecting the objectively best option, our study focuses on decision consistency as the important criterion of decision quality. In other words, we want to investigate how consistent decision-makers are in sticking to their stated preferences when exposed to various amounts of information. This angle is critical to add to the literature regarding the impact of information volume in distorting human subjects' decision criteria. The resulting findings not only contribute to academic knowledge but also offer practical guidance for effective real-world decision making under information overload, as well as practical suggestions for information disclosure in decision system designs. This insight bridges behavioural decision-making and interface design, offering a mechanism by which operational systems can scaffold user attention and mitigate preference distortion.

2.2. Inconsistency between stated preference and revealed preference

While research on hypothetical bias focuses on mismatches between what people say in hypothetical scenarios and what they do when real stakes are involved, our study investigates a parallel inconsistency within hypothetical-but-structured settings. Preferences can be either stated or revealed

through choices and actions. Rationally speaking, given a certain preference, these two elicitation methods should yield the same outcome and be consistent. However, empirical studies have revealed the opposite. Existing research primarily focuses on and has extensively documented hypothetical bias, noting that what people report in hypothetical scenarios often deviates from their actions in real-world situations with real consequences (Fifer et al. 2014, Quaife et al. 2018). The hypothetical bias manifested across different fields, such as healthcare, psychology, and management. Özdemir et al. (2009) found that people tend to overestimate their willingness to pay (WTP) for medical treatment. Participants in the control group had significantly higher WTP than those who were reminded of the real-world cost by a cheap-talk strategy that reduced their hypothetical bias. In a more recent study, researchers found that people reported their frequency of blood donation as 30% to 41% higher than their actual donation frequency (de Corte et al. 2021). In management settings, Verma and Pullman (1998) showed that although managers stated that they perceived quality as the most important attribute in supplier selection, they actually based their decisions more frequently on price and delivery time. The primary cause of hypothetical bias is the discrepancy in monetary consequences between hypothetical and real scenarios. In hypothetical scenarios where monetary consequences are absent, people tend to express a greater preference for choices associated with a positive social image and morality (Johansson-Stenman and Svedsäter 2012). Even though these options often involve higher actual monetary costs, individuals do not weigh these costs heavily because they do not pay the cost. Conversely, in real scenarios where monetary consequences are present, people show a stronger preference for options that incur lower actual costs. This research contributes to the literature on stated versus revealed preferences by introducing a novel form of inconsistency—divergence in attribute weighting—under controlled cognitive load conditions. First, unlike prior studies focusing on preference inconsistency in choices or willingness to pay (WTP), we explore a novel type of preference inconsistency at the attribute level. Specifically, we examine how individuals rank the importance of each attribute (i.e., stated preference) and how this ranking differs from the actual weight they assign to each attribute when choosing a supplier. This type of inconsistency not only differs from hypothetical bias in terms of the phenomena observed but also in the underlying mechanisms. In the context of our study, we keep monetary consequences constant for stated preference and revealed preference, and factors typically associated with moral goods or social image are absent. These approaches ensure that neither monetary consequences nor social desirability can explain the discrepancy we identify. Second, we investigate how information volume could aggravate preference inconsistency. In our research, we investigate the impact of information volume and information disclosure sequence on SP-RP inconsistency in decision making. The resulting findings can guide managers to be aware of decision distortion due to information volume. Our research also suggests that in situations

requiring users to process large amounts of information, platforms use a progressive sequencing of information disclosure to reduce the degree of SP-RP inconsistency in decision making. By controlling for social desirability and incentive salience, we demonstrate that inconsistency can arise purely from interface complexity, thus offering a distinct behavioural mechanism.

2.3. Supplier Selection

The setting of this study closely connects to procurement literature. Related empirical studies mainly focus on how procurement methods and contract terms affect the performance of the contracted supplier. Specifically, researchers have investigated the impact of choice between auctions versus negotiations (Bajari et al. 2008), of contractual completeness on long-term projects (Crocker and Reynolds 1993), of contract type on product reliability (Guajardo et al. 2012), of government oversight contingencies on contract value (Calvo et al. 2019), and of internet use during search and purchase processes (Mishra et al. 2007). Our research shifts the focus to how interface-level information disclosure affects preference articulation during the supplier selection stage, a topic underexplored in the procurement literature. As we identify the impact of information volume on retailers' decision-making processes, we propose a progressive information disclosure to improve retailers' adherence to their stated preferences in actual decisions.

Another stream of related literature in procurement studies is about the decision-making process for selecting suppliers. The purpose of this stream of research is to select an optimal supplier based on the retailer's known preference for certain supplier attributes. Existing literature has utilized methods including constrained optimization (Hong et al. 2005), multi-objective programming (Weber and Ellram 1993, Narasimhan et al. 2006), analytic hierarchy process (Akarte et al. 2001, Chan and Kumar 2007), and data envelopment analysis (Weber and Desai 1996, Talluri et al. 2006). Our research differs as the goal is to identify the discrepancy between retailers' stated preferences and their actual revealed preferences and investigate information volume's role in this inconsistency. Whereas prior studies assume known and stable preferences in applying supplier selection models, we highlight how interface design itself can shape—if not distort—those preferences before such models are applied.

Building on the literature on bounded rationality and information overload, we propose that higher information complexity increases cognitive burden, disrupting preference stability. We further hypothesise that exposing decision-makers first to simplified information environments enables them to anchor their attribute evaluations, thus supporting consistency under later complexity. Together, we propose the following hypotheses:

Hypothesis 1. Retail customers are more likely to demonstrate inconsistency between their stated preferences and revealed supplier choices when evaluating options described by a high number of attributes, as compared to options with fewer attributes.

Hypothesis 2. A progressive transition from low to high information volume—starting with a smaller set of attributes—reduces decision inconsistency by helping decision-makers stabilize their attribute weighting early in the process.

These hypotheses aim to uncover not only the distorting role of information volume on decision-making, but also the corrective potential of interface sequencing—a design lever with significant implications for digital procurement systems.

3. Experiments Design

To test our hypotheses under controlled conditions, we designed a series of online lab experiments that manipulate information volume and presentation sequence in supplier selection tasks. We conducted three sets of experiments to study the impact of information volume on decision inconsistency.

3.1. Study 1: Short Version vs. Long Version

In this study, we investigate whether retailers allocate weights distinctively on supplier attributes when the number of supplier attributes is high. We examine whether participants' supplier selections reflect their self-reported attribute importance levels, and whether this consistency varies systematically with the volume of information presented, thereby testing Hypothesis 1.

We conducted this first laboratory study on Prolific and recruited 309 participants. We informed them that the experiment would last approximately 10 minutes, with an average compensation of $1.3. To incentivize participants to make consistent supplier selection decisions, we offered a base compensation of $1.3, with the potential to earn an additional $1 bonus if their supplier selection decisions aligned with their reported attribute importance rankings. The incentive was designed to maintain participant engagement and minimize data noise resulting from inattention and low motivation.

The laboratory experiment is designed to mimic the supplier selection decisions that sourcing managers face in the B2B online environment. We focus on the fashion industry as our study context. Participants are tasked with selecting suppliers to produce a specified quantity of clothing products. At the start of the experiment, participants are introduced to key supplier attributes (available in our online supplement). Following the tutorial, we test participants' comprehension of the supplier attributes. Participants are required to categorize the list of supplier attributes into three dimensions: quality, speed, and service. They must achieve 100% accuracy in this classification test before moving to the task stage.

Participants are required to complete two sets of tasks. First, they are asked to state importance rankings for supplier attributes by rating each attribute on a 1-5 scale, from "not at all important" to "critical". After participants have self-reported their importance rankings, they will complete

Figure 2 Experiment Procedure

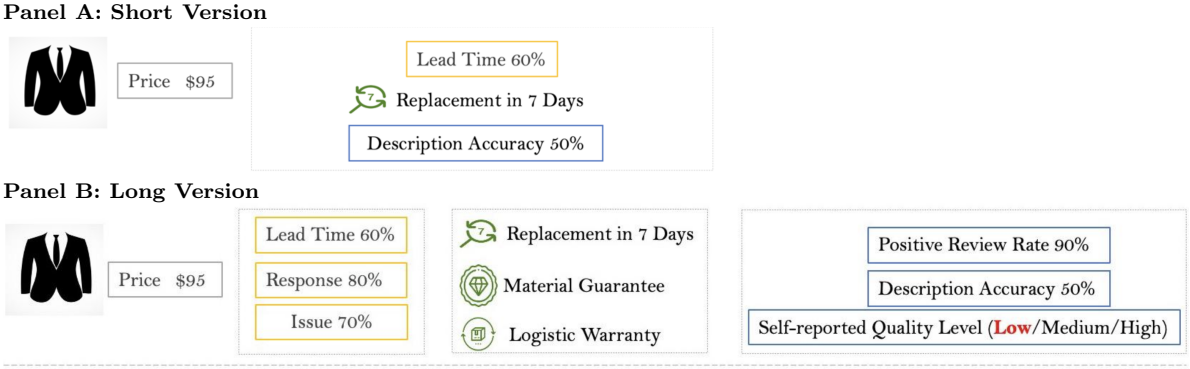


a set of six supplier selection tasks. The instruction page that connects the two sets of tasks also emphasizes the goal of decision consistency and highlights that participants can earn $1 bonus compensation if their reported importance aligns with their supplier selection choices. In each supplier selection task, participants must select one supplier out of three candidates by evaluating the trade-offs among the supplier attributes. Note that, by design, there is no Pareto optimal supplier for any task. During this selection process, participants act on and thus implicitly reveal their actual preferences across supplier attributes. After both sets of tasks, we perform a sanity check to assess participants' understanding of the attributes and collect demographic information. The experimental procedure is outlined in Figure 2.

Participants are equally and randomly assigned to one of three experiment conditions: long version, short version 1, and short version 2. In the long-version condition, in each supplier selection task, we list the value of ten attributes of the three candidate suppliers. In the short-version conditions, the number of supplier attributes reduces to four. We select two different subsets of attributes to create the two short versions. Specifically, price is a constant attribute, and we randomly select one attribute for each dimension: quality, speed, and service. We discuss the attribute choice in detail in Section 4.2 and illustrate the sample profile of candidate suppliers by long and short versions in Figure 3.

Given that we adopted a similar design across all three studies, we did not pre-register. We apply predetermined exclusion criteria to filter out low-quality responses. Specifically, the last supplier selection question includes a Pareto inferior candidate supplier; participants are excluded if they

Figure 3 Sample Profile of Candidate Suppliers



choose this inferior supplier. Our final dataset includes responses from 301 participants, with 98 completing the long-version experiment and 104 and 99 participating in the two short-version experiments. Among all 301 participants, we provide bonus compensation to 116 participants. We determine the bonus compensation again through the last supplier selection question. Other than the Pareto inferior candidate supplier, we characterize the other two supplier candidates such that they only differ in two of their listed attributes. We assign the bonus compensation based on whether participants' supplier choices align with their reported ratings between the two attributes. This design allows us to isolate the effect of attribute volume on SP-RP inconsistency while holding constant the number of supplier options, thus disentangling attribute-level overload from choice overload.

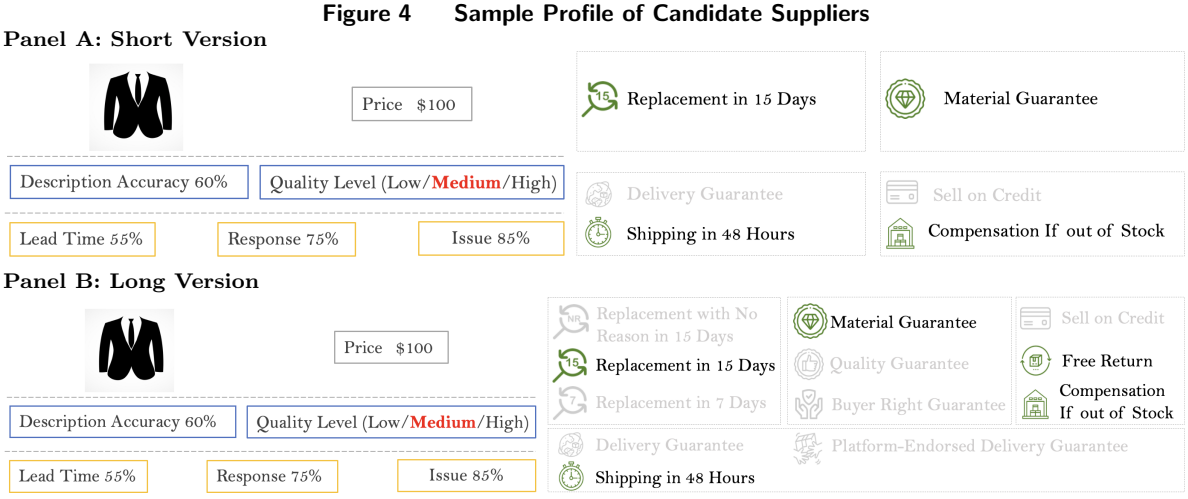
3.2. Study 2: Short-to-Long Version versus All Long Version

Study 2 explores whether variations in decision consistency, based on the number of supplier attributes, persist in subsequent supplier selection decisions (H2). To investigate this, Study 2 utilizes two between-subjects conditions: short-to-long versus all-long. In both conditions, participants complete two stages of tasks: the second stage always features a long version task, while the first stage includes a short version task in the short-to-long condition and a long version task in the all-long condition. If early exposure to simplified attribute sets facilitates the stabilisation of attribute weighting, we expect participants in the short-to-long condition to demonstrate higher decision consistency in the second-stage long-format tasks compared to those in the all-long condition.

Study 2 recruited 200 participants on Prolific. Similar to the design of Study 1, we informed participants that the experiment would take approximately 15 minutes. We offer a base compensation of \$2.3 upon completing the experiment and provide an additional \$1 bonus compensation to incentivize participants to make aligned supplier selection decisions.

We largely follow the experimental setup of Study 1, with the only modification being the design of the supplier selection tasks. In this second experiment, after participants rate the importance of supplier attributes, they are required to complete a total of 12 supplier selection tasks. As mentioned before, participants are randomly assigned to one of two experiment designs: the short-to-long version or the all-long version. In the short-to-long version, participants first complete six short-version supplier selection tasks, where we describe each candidate supplier with four attributes (as shown in Panel A of Figure 3). They then complete six long-version tasks, where candidate suppliers are characterized by ten attributes (as shown in Panel B of Figure 3). In the all-long version, participants are required to complete 12 long-version tasks. We focus on participants' decision consistency in the last six long-version tasks to determine whether starting with short-version tasks, as opposed to long-version tasks, would influence the consistency of subsequent

Figure 4 Sample Profile of Candidate Suppliers



long-version supplier selection decisions. Note that the last six long-version tasks are identical in both experiment designs to ensure a fair comparison.

After excluding ineligible participants using the same filtering question described in Study 1, we observed 96 participants in the short-to-long group and another 95 in the long-to-long group. Across both groups, 60 participants received bonus compensation that honored the decision's consistency. Importantly, the long-version tasks in the second stage are identical across both conditions, enabling us to cleanly identify the effect of initial exposure on subsequent decision consistency.

3.3. Study 3: Short-to-Long Version versus Long-to-Short Version

Study 3 was a conceptual replication of Studies 1 and 2. In this third experiment, we recruited 162 participants (46.91% male; 51.85% female; 1.23% prefer not to disclose) from a U.S. university. The participants are undergraduate students majoring in business. They have learned about supplier selection in an operations management class and choose to complete the experiment for the compensation of extra credit in coursework. This conceptual replication tests the robustness of our findings across variations in participant population, attribute set size, and the sequence of preference elicitation and choice tasks.

Participants were equally and randomly assigned to one of two experiment conditions that varied by the order of viewing long and short versions of supplier selection tasks. In total, 80 participants completed the long-to-short experiment, and 82 participants completed the short-to-long experiment. Participants in the short-to-long design group first view six short-version supplier selection tasks with 12 attributes to describe each supplier (as illustrated in Panel A of Figure 4). They then view six long-version tasks that list 18 attributes (as illustrated in Panel B of Figure 4). In the long-to-short group, participants first view the six long-version supplier selection tasks, followed by six short-version ones. Appendix Table 8 summarizes the list of all supplier attributes used in Study 3 and whether or not they appear in the short version.

We also performed a trial experiment to test whether the presentation format might influence participants' choices. Specifically, we compared the supplier selection results from experiments conducted via a grid view and a list view and found no difference in choices at our metrics of interest. Hence, we used the grid view for this third experiment, as shown in Figure 4. This choice also allowed us to control for presentation effects, ensuring that observed inconsistency patterns stem from information volume and sequence, rather than visual display format. Also, to ensure participants paid enough attention to the educational information and exerted thought when selecting suppliers and answering our questions, they would need to stay on each supplier selection page for at least 15 seconds before proceeding to the next task.

4. Empirical Method

In this section, we first describe the empirical model we use to estimate the effect of supplier attributes on supplier selection decisions. We then introduce how we cluster and standardize the supplier attributes for empirical identification.

4.1. Model Specification

The outcomes of the supplier selection tasks reveal the preferences of participants. Assume that, for each task, the participants select the supplier that provides them the highest utility; we define the following conditional logit model to estimate the effect of supplier attributes on selection decisions. The variations in the supplier attributes drive the supplier selection choices, and we thus express the choice utility as a function of the supplier attributes:

$$U_{ij} = X_j^T \cdot \beta + \epsilon_{ij}, \tag{1}$$

where U_{ij} denotes the utility for participant i to choose supplier alternative j. Vector X_j includes the supplier attributes introduced in Table 8. ϵ_{ij} is the unobserved random utility shock that follows the Gumbel distribution. The main outputs of the model are the sets of estimates $\{\beta\}$, generated from fitting the equation using various sets of supplier selection tasks. We expect to observe a negative correlation between the price and the choice utility. For all other supplier attributes, we expect to observe positive correlations with the choice utility. We use the maximum likelihood method to obtain the estimates from the joint probability distribution. We cluster the standard errors at the participant level to further control for potential correlations in error terms.

4.2. Attribute Choice and Attribute Clustering

As one of the most critical activities in the sourcing process, the supplier selection issue has been extensively studied in the operations management community. The modeling literature investigates how retailers can use mechanism designs (Weber and Ellram 1993) and selection algorithms (Talluri

Table 1 Supplier Attributes

Dimension	Attributes
Price	Price
Quality	Self-Report Quality Level, Description Accuracy, Quality Rating
Speed	Lead Time, Response Rate, Issue Rate
Service	Replacement in 7 Days, Material Guarantee, Logistic Warranty

2002) to find the optimal supplier by comparing suppliers' price, product quality, order-to-delivery speed, and ancillary services. This choice of dimensions is supported by the survey literature, which unanimously finds that quality, delivery performance, and price are the three most important factors in supplier selection (i.e., Patton III 1996 and Kar and Pani 2014). The wave of servitization in traditional manufacturing industries promotes the importance of the service factor in current society (Cohen 2012). Specifically, Guajardo et al. (2012) empirically show that the service dimension plays an important role in the supplier selection process and can compensate for inferior attributes in other aspects.

Following the supplier selection literature, we categorize supplier attributes into four dimensions: price, quality, speed, and service. Table 1 outlines the ten supplier attributes used in Study 1 and Study 2. We select three distinct attributes for each of the quality, speed, and service dimensions, then combine them with price to create the ten attributes for the long-version experiments. For the short-version experiments, we select one attribute from each of the quality, speed, and service dimensions, along with price, to create a list of four attributes. To ensure that our findings are not dependent on the specific attributes chosen, we use two different subsets of the long-version supplier attributes to create two short-version experiments.

The participants' importance ratings of each attribute on a 1-5 scale reflect their underlying preferences for the different supplier attributes. For rational participants, the self-reported preferences should match their weight on the attributes for the supplier selection tasks. We rely on the sets of estimates $\{\beta\}$ generated from Equation 1 to evaluate whether participants make supplier selection decisions following their self-reported importance levels. We mainly focus on the rankings of attributes and define consistency as whether the importance rankings retrieved from the estimates reflect the rankings suggested by the self-reported importance levels. To ensure that the estimates are comparable across attributes, we standardize the value of each of the attributes to have a mean of 0 and a standard deviation of 1.

We recognize that the number of supplier attributes affects the ranking correlations. To minimize the influence of varying attribute sizes on our findings, we also perform attribute clustering in the long-version experiments across the quality, speed, and service dimensions to create a comparable

four-attribute scenario, similar to the short-version experiments, for our empirical analysis. Specifically, for each candidate supplier, we average the three standardized attributes within the quality, speed, and service dimensions to create a single factor representing each dimension.

In Study 3, the list of supplier attributes is slightly modified, with a greater number of attributes included in both the short and long versions. For brevity, we discuss the attribute clustering and standardization for Study 3 separately in Appendix Section 4.2.

5. Empirical Results

This section presents the empirical results that demonstrate the inconsistency between self-reported and actual importance rankings in supplier selection decisions. First, we discuss how this inconsistency varies across task versions. Next, we explore approaches to mitigate decision inconsistency in long-version tasks, followed by robustness check analyses.

5.1. The Impact of Information Volume on Decision Inconsistency

Based on the experiment results from Study 1, we present in Table 2 the participants' reported importance rating for the list of supplier attributes and the estimation results from their supplier selection decisions using the conditional logit model. To allow a direct comparison between the short-version and long-version tasks, we report the aggregated estimates and average ratings for the long-version. We present the alternative long-version results using all ten supplier attributes in Appendix Table 3.

The estimates in Table 2 indicate an inconsistency between participants' self-reported importance ratings and their revealed preferences during supplier selection. For example, considering all tasks, in the long-version group, participants' self-reported importance for the speed dimension is moderate (average rating of 3.517), but the revealed preference suggests a negative alignment. These discrepancies suggest that high information volume disrupts individuals' ability to act in accordance with their stated preferences, consistent with cognitive overload and bounded attention theories. In the short-version group, when comparing the rankings between the self-reported ratings and the choice model estimates, we observe that the speed and service dimensions have switched places for participants.

Based on the observations, we now generate metrics to quantify and compare the consistency in the participants' self-reported importance ratings (SR) and the estimated actual importance level of supplier attributes (RP) from the supplier selection choices. For each pair of supplier attributes, we perform a two-sided, two-sample t-test to compare the self-reported importance ratings between each possible pair of supplier attributes. Similarly, we conducted the chi-square test to compare the estimated coefficients and identify the actual importance the participants assigned to each supplier attribute in their selection tasks. Because price is the only supplier attribute expected

Table 2 Study 1: Effect of Supplier Attributes on Selection Decisions

Study 1: Short vs. Long	Choice Model Estimation All Tasks		Choice Model Estimation Second Half		Self-Report Importance Level	
Attributes	Short (1)	Long (2)	Short (3)	Long (4)	Short (5)	Long (6)
Price	-3.144***	-1.989***	-7.007***	-3.137***	4.286	4.184
	(0.411)	(0.476)	(0.934)	(0.737)	(0.794)	(0.912)
Quality	1.182***	1.226***	1.775***	1.762***	4.049	3.946
	(0.067)	(0.106)	(0.159)	(0.439)	(0.963)	(0.566)
Speed	0.292***	-0.169*	0.913***	0.247	3.527	3.527
	(0.063)	(0.088)	(0.151)	(0.501)	(0.946)	(0.660)
Service	0.588***	0.598***	0.747***	1.044***	3.148	3.816
	(0.058)	(0.064)	(0.082)	(0.222)	(1.033)	(0.835)
Participants	203	98	203	98	203	98
Pseudo R2	0.3253	0.2920	0.4154	0.3803		
Observations	3,654	1,764	1,827	882		

Note. We report the coefficient estimates from conditional logit models in columns (1) to (4). We include all six tasks in the regression to generate estimates in columns (1) and (2). We then focus on the second half of tasks, i.e., the last three questions, to generate estimates in columns (3) and (4). We present the clustered standard errors at the participant level in parentheses. We report the average and the standard deviation of participants' reported importance rankings in columns (5) and (6). *** — 0.01 level, ** — 0.05 level, and * — 0.1 level..

to bring down the choice probability, we flip the sign of the estimated price effect and focus on the difference in the absolute value of estimated coefficients. We summarize the inconsistency in Table 3 and provide the detailed pairwise comparison results for both long and short-version tasks in Appendix Table 4.

Note that the number of supplier attributes varies between short-version and long-version tasks, resulting in a difference in the total number of comparable supplier attribute pairs. With ten supplier attributes in the long-version group, we need to consider 45 comparable supplier attribute pairs. In comparison, the four supplier attributes in the short-version group generate six comparable pairs. For each comparable pair, we compare the statistical test results between the self-reported and the estimated supplier importance. We label a comparison pair as inconsistent if the two test results (i.e., $>$, $=$, or $<$) do not match. This pairwise comparison approach enables a nuanced evaluation of attribute-level discrepancies beyond aggregate model fit, offering insight into the micro-level structure of inconsistency. We consider two p-value thresholds for the pairwise comparison, 0.05 and 0.1.

We count the total number of inconsistent comparison pairs for both short-version and long-version groups and report the results in Table 3. To ensure a fair comparison between task versions, we report the count of inconsistent pairs for the long-version group based on 1) the clustered attributes and 2) the individual supplier attributes. We also present the percentage of inconsistent rankings for easy reference.

Table 3 Study 1: Inconsistency between Self-Reported and Estimated Actual Importance Rankings

Study 1: Short vs. Long	All Tasks						Second Half			
	Short		Long				Short		Long	
Attributes	Individual		Cluster		Individual		Individual		Cluster	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: p-Value Threshold = 0.05										
No. (Pct.) of Inconsistent Rankings	1	(16.7%)	2	(33.3%)	24	(53.3%)	1	(16.7%)	1	(16.7%)
No. (Pct.) of Flipped Estimation Signs	0	(0%)	0	(0%)	0	(0%)	0	(0%)	0	(0%)
Total No. of Pairs	6		6		45		6		6	
Panel B: p-Value Threshold = 0.1										
No. (Pct.) of Inconsistent Rankings	1	(16.7%)	1	(16.7%)	24	(53.3%)	1	(16.7%)	1	(16.7%)
No. (Pct.) of Flipped Estimation Signs	0	(0%)	1	(16.7%)	0	(0%)	0	(0%)	0	(0%)
Total No. of Pairs	6		6		45		6		6	
Ranking Correlation	0.8		1		0.079		1		1	

Note. We report the number of inconsistent comparison pairs in odd columns and the relative percentage in parentheses in even columns. When we focus on the second half of the tasks, i.e., the last three questions, we cannot estimate the choice model with individual questions in the long version due to the limited sample.

Table 3 demonstrates inconsistency between the self-reported ratings and the estimated attribute importance levels from the choice model. This inconsistency is particularly pronounced in the long-version group. The negative estimated impact of speed factors on selection outcomes is particularly concerning. In addition, participants in the long-version group place only a marginally higher weight on price than quality, despite reporting significantly higher ratings for the price factor. Additionally, participants focus more on quality attributes than service ones in supplier selection tasks, despite their self-reported ratings suggesting minimal differences.

Furthermore, by comparing the inconsistency metrics across all tasks and the second half of the tasks, we observe that participants in both the short- and long-version groups improve their consistency performance over time. To better understand the underlying mechanisms of this learning effect, we examine the changes in the decision time. Our analysis suggests that the performance improvement in the long-version group can be attributed to the increased decision time spent on the second half of the questions. Specifically, participants in the long-version group spent 74.1% more time per question (= 19.62 sec/11.27 sec - 1) in the second half of the survey, whereas participants in the short-version group only increased their time per question by 15.1% (= 9.28 sec/8.06 sec - 1).

5.2. Inconsistency Carryover In the Transitional Information Display

We analyze the results from Study 2 to test the carry-over conjecture and examine whether the prior task version affects participants' behavior in subsequent supplier selection decisions. As such, in the conditional logit choice model, we focus on decision inconsistency in the last six long-version questions under both experimental formats: short-to-long and all-long.

Table 4 Study 2: Inconsistency between Self-Reported and Estimated Actual Importance Rankings

Study 2:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Survey Format: SL				Survey Format: LL			
Attributes	Cluster		Individual		Cluster		Individual	
Panel A: p-Value Threshold = 0.05								
No. (Pct.) of Inconsistent Rankings	1	(16.7%)	26	(57.8%)	2	(33.3%)	26	(57.8%)
Total No. of Pairs	6		45		6		45	
Panel B: p-Value Threshold = 0.1								
No. (Pct.) of Inconsistent Rankings	1	(16.7%)	27	(60%)	2	(33.3%)	27	(60%)
Total No. of Pairs	6		45		6		45	
Ranking Correlation			0.248				0.188	
Top 60% Attributes Ranking Correlation			0.6				-0.03	

Note. We report the number of inconsistent observations in odd columns and the relative percentage in parentheses in even columns.

Table 4 summarizes the participants' decision inconsistency performance between the two experiment formats. The associated full-choice model estimation results are included in Appendix Table 5. Both formats show similar inconsistency levels when comparing the number of inconsistent pairs for individual attributes. However, after aggregating quality, speed, and service attributes into their respective clusters, participants in the all-long group exhibit a higher level of inconsistency in their second set of six supplier selection decisions. This supports the behavioural interpretation that initial exposure to complexity impairs the formation of consistent selection rules, whereas a scaffolded progression from simple to complex mitigates that effect. More importantly, when comparing the correlation between self-reported ranking and estimated importance ranking for the ten supplier attributes, participants in the short-to-long group outperform in both overall and top attributes ranking correlations. These findings support the hypothesis that early exposure to simplified information environments facilitates more stable attribute weighting, which persists into more complex decision settings.

To further investigate the task sequence's effect on inconsistency in supplier attribute selection, we assign participants to short-to-long and long-to-short groups in Study 3. Table 5 presents the sub-sample estimation results by task versions and the display sequence. Comparing the estimates in columns (1) and (2) and in columns (3) and (4), we can see that the supplier selection criteria differ by the task sequence. For both long and short-version tasks, the estimates change significantly depending on whether the tasks are offered to the participants as the first set or the second set of questions. The observation that the sequence of experiment tasks affects the estimated effects of supplier attributes further justifies our decision to investigate the potential carryovers of supplier selection criteria from the first set to the second set of questions.

Comparing the estimates in columns (5) and (6) and in columns (7) and (8), we note that participants tend to assign similar weights to supplier attributes regardless of the task version. In

Table 5 Study 3: Effect of Learning in Selection Decisions

Study 3: Choice Model	(1) Long Version First Set	(2) Long Version Second Set	(3) Short Version First Set	(4) Short Version Second Set	(5) Survey Format: LS Long - 1	(6) Survey Format: LS Short - 2	(7) Survey Format: SL Short - 1	(8) Survey Format: SL Long - 2
Price	-0.040	-0.497***	-0.379***	-0.062	-0.040	-0.062	-0.379***	-0.497***
	(0.113)	(0.123)	(0.130)	(0.122)	(0.113)	(0.122)	(0.130)	(0.123)
Quality (Combined)	0.528***	0.850***	0.534***	0.368**	0.528***	0.368**	0.534***	0.850***
	(0.161)	(0.134)	(0.116)	(0.161)	(0.161)	(0.161)	(0.116)	(0.134)
Speed	0.210**	0.315***	0.442***	0.025	0.210**	0.025	0.442***	0.315***
	(0.089)	(0.083)	(0.078)	(0.058)	(0.089)	(0.058)	(0.078)	(0.083)
Guarantee Service	1.012***	0.720***	0.513***	0.192***	1.012***	0.192***	0.513***	0.720***
	(0.173)	(0.126)	(0.065)	(0.051)	(0.173)	(0.051)	(0.065)	(0.126)
Delivery Service	0.457***	0.135	0.488***	0.360***	0.457***	0.360***	0.488***	0.135
	(0.075)	(0.117)	(0.092)	(0.057)	(0.075)	(0.057)	(0.092)	(0.117)
Replacement Service	0.381***	0.228**			0.381***			0.228**
	(0.078)	(0.111)			(0.078)			(0.111)
Additional Service	0.349***	0.481***	0.329***	0.434***	0.349***	0.434***	0.329***	0.481***
	(0.076)	(0.081)	(0.058)	(0.059)	(0.076)	(0.059)	(0.058)	(0.081)
Participants	80	82	82	80	80	80	82	82
Observations	1,440	1,476	1,476	1,440	1,440	1,440	1,476	1,476
Pseudo R2	0.179	0.265	0.210	0.162	0.179	0.162	0.210	0.265

*Note.*In the first four columns, we arrange the results first by the version type (long or short) and then by the sequence of question displays. In the last four columns, we arrange the results first by the sequence and then by the version type. We report the coefficient estimates from conditional logit models. We present the clustered standard errors at the participant level in the parentheses. *** — 0.01 level, ** — 0.05 level, and * — 0.1 level.

other words, the participants likely settle into their supplier selection criteria with the first set of tasks. The weights they assign to each supplier attribute might be carried to their second set of tasks. Table 6 provides support for this conjecture. Comparing columns (1) and (3) and columns (5) and (7), we can see that the number of inconsistent comparison pairs is largely similar between the two sets of tasks within each experiment format. As depicted in columns (1)–(2), there is generally a high level of inconsistency for those participants who start with the long version tasks.

Conversely, as shown in columns (3) and (5), the inconsistency is smaller if participants start with the short version tasks. More importantly, we observe participants could carry this low level of inconsistency to their later set of long-version tasks, resulting in significantly more consistent supplier selection decisions. These results suggest that decision-makers are more likely to reflect their true underlying preferences if there is a progressive display of attributes, where the platform begins with shorter and ends with longer amounts of information. The online B2B sourcing platforms have abundant information available for each supplier candidate, requiring extended decision making from customers. To facilitate better sourcing decisions, platforms might consider a supplier comparison interface that provides a short initial list of key supplier attributes and allows users to include additional attributes as needed.

Table 6 Study 3: Inconsistency between Self-Reported and Estimated Actual Importance Rankings

Study (3)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Survey Format: LS				Survey Format: SL			
	Long - 1		Short - 2		Short - 1		Long - 2	
Panel A: p-Value Threshold = 0.05								
No. (Pct.) of Inconsistent Rankings	15	(71%)	14	(93%)	12	(80%)	12	(57%)
No. (Pct.) of Rankings with Flipped Signs	5	(24%)	6	(40%)	0	(0%)	1	(5%)
Total No. of Pairs	21		15		15		21	
Panel B: p-Value Threshold = 0.1								
No. (Pct.) of Inconsistent Rankings	14	(67%)	14	(93%)	13	(87%)	12	(57%)
No. (Pct.) of Rankings with Flipped Signs	5	(24%)	6	(40%)	0	(0%)	1	(5%)
Total No. of Pairs	21		15		15		21	

Note. We report the number of inconsistent comparison pairs in odd columns and the relative percentage of inconsistent rankings in the parentheses in even columns.

5.3. Robustness Checks

We now examine whether our results can be explained by collinearity when we include a long list of supplier attributes. When we designed the experiment, we strategically created supplier candidates for each supplier selection task to ensure there were no Pareto optimal choices. As a result, supplier candidates at low price points tend to have worse quality performance, slower delivery speed, and fewer service offerings. In particular, the correlation between price and quality attributes is relatively high. Consistent with prior supplier selection literature, participants disproportionately weight price and quality in their revealed preferences, making these dimensions key drivers of SP-RP inconsistency under overload. To alleviate the potential impact of this collinearity on the estimation of conditional logit models, we create a combined price and quality factor by aggregating the normalized price and quality factors.

The sign of the normalized price factor is flipped before the aggregation to reflect that a supplier candidate with lower prices is more attractive to decision-makers. We also perform a second standardization to ensure that the combined factor has a zero mean and unit standard deviation. In Panels A and B of Table 7, we observe robust findings about the smaller inconsistency levels in short-version tasks. The detailed estimation results from using this combined price and quality factor are included in Appendix Table 6 and 10.

We require participants to work on additional problems in Study 3 to determine their decision-making styles. Correspondingly, we categorize our participants into two groups, i.e., satisficers and maximizers. The decision science literature (Schwartz et al. 2002, Bloch et al. 2003) has identified that decision-makers' styles significantly affect how they perceive and process abundant options or extensive amounts of information. The satisficers tend to choose the first option above her/his threshold, while the maximizers want to find the best choice among all candidates. We adopt the method used in Townsend and Kahn (2014) to recover the participants' decision-making styles.

Table 7 Robustness Checks

Study 2	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Survey Format: SL				Survey Format: LL			
	Cluster		Individual		Cluster		Individual	
Panel A: Combine Price and Quality Factors								
No. (Pct.) of Inconsistent Comparisons	1	(33.3%)	10	(47.6%)	1	(33.3%)	12	(57.1%)
No. (Pct.) of Rankings with Flipped Signs	0	(0%)	3	(14.3%)	0	(0%)	5	(23.8%)
Total No. of Pairs		3		21		3		21
Ranking Correlation				0.25				0.143
Top 60% Attributes Ranking Correlation				0.6				0.6
Study 3	Survey Format: LS				Survey Format: SL			
	Long - 1		Short - 2		Short - 1		Long - 2	
Panel B: Combine Price and Quality Factors								
No. (Pct.) of Inconsistent Rankings	12	(80%)	9	(90%)	8	(80%)	7	(47%)
No. (Pct.) of Rankings with Flipped Signs	4	(27%)	6	(60%)	0	(0%)	0	(0%)
Total No. of Pairs		15		10		10		15
Panel C: Satisficers								
No. (Pct.) of Inconsistent Rankings	18	(86%)	13	(87%)	13	(87%)	13	(62%)
No. (Pct.) of Rankings with Flipped Signs	2	(10%)	2	(13%)	0	(0%)	0	(0%)
Total No. of Pairs		21		15		15		21
Panel D: Maximizers								
No. (Pct.) of Inconsistent Rankings	15	(71%)	13	(87%)	11	(73%)	9	(43%)
No. (Pct.) of Rankings with Flipped Signs	4	(19%)	4	(27%)	0	(0%)	0	(0%)
Total No. of Pairs		21		15		15		21

Note. We report the number of inconsistent rankings in odd columns and report the percentage of inconsistent rankings in the parentheses in even columns. p-Value Threshold = 0.05.

We then calculate the total score and assign the participants to the satisficer or the maximizer group based on the median split. Table 7 demonstrates the estimation results for the two groups from the conditional logit model. We can tell from the summarized inconsistency levels in Panels C and D that both styles of decision-makers are subject to similar levels of decision inconsistency. Taken together, these robustness checks reinforce that the observed SP-RP inconsistency is not merely a statistical artefact of model collinearity or task fatigue, but instead reflects a systematic behavioural pattern shaped by interface design.

6. Managerial Implications

In our research, we investigate how information volume impacts the inconsistency between stated and revealed preferences in the human decision-making process. First, by varying the degree of information or attributes a person is faced with when making decisions, we discover that inconsistency between stated and revealed preferences can be mitigated through progressive information disclosure. To improve preference alignment in supplier selection, platforms may implement progressive information disclosure—starting with a focused subset of attributes and offering optional

access to additional information. By reducing the gap between users' stated priorities and their final selections, such interface designs can lead to improved supplier matches, greater customer satisfaction, and more reliable downstream procurement outcomes.

Furthermore, our research confirms that a person's revealed preference that he/she actually uses in the decision-making process (the allocated weights on attributes) does not necessarily coincide with his/her stated preference. Our experiment results show that it is not the information volume alone but also the sequence for displaying the information that matters in determining the degree of inconsistency. Our experiments suggest that progressive information disclosure reduces cognitive overload and enhances consistency between stated and revealed preferences—providing a low-cost, implementable design lever for platform operators. In the following, we discuss several specific examples to demonstrate how user interface platforms can implement progressive information disclosure to facilitate consistent decision-making criteria.

The first example comes from the setting of search-facilitated online platforms. When customers search keywords, an online platform attentive to mitigate the information overload phenomenon would first release information only about key attributes of candidate choices on the main search page, then reveal more attribute information on subsequent detailed pages once a customer clicks on them. In China, the restaurant review platform *Meituan* now returns search results with only a few keywords characterizing each listed restaurant. This is more advanced than it may at first appear, since the listed keywords are generated in a customized way to provide the information that the searching person most likely cares about finding based on his/her search keywords. For example, if a person searches "family reunion dining," then the returned restaurants' characterizing keywords will be focused on whether the dining environment or menus are suitable for family dining. After a person clicks on the detailed page of a specific restaurant, he/she can obtain more comprehensive information regarding the restaurant. In our setting, we suggest that the supplier selection platform list a few key attributes of the supplier's information in the search list, then offer additional pages with more comprehensive information disclosure for the retailers to further specify their initial choices. With such progressive information disclosure, the online platform can reduce the inconsistency of its users' decisions between their stated and revealed preferences.

Another example comes from the e-commerce setting. In certain B2C situations, such as when shopping for electronics or cruise packages, customers are likely to face an overwhelming amount of information characterizing the attributes of the products or services. We suggest that both B2C and B2B e-commerce platforms offer products and services using a user interface design that does not initially list all the attributes. That is, e-commerce websites would do well to offer customers an opportunity to pick a few key variables to initially organize information and generate optional additional information, such as a comparison table. Such platforms could allow users to expand

attribute views incrementally (e.g., via 'see more attributes' options), preserving cognitive focus while still offering comprehensive evaluation tools when needed. With a user interface design that progressively provides information, customers can make decisions based on variables they most care about, consistent with their stated preferences, and thus reduce the degree of information distortion in decision criteria. In practical terms, these findings imply that e-commerce and procurement platforms need not personalise attribute display immediately; a default progressive design can already produce more consistent decision behaviour across user segments. We offer several exciting topics for future study following our research. The first would be to further investigate how different types of information volume may impact human decision-making inconsistency. Information volume can occur in two ways in decisions: one is the number of choices; the other is the number of attributes that characterize each choice. In our research, we hold constant the number of choices and investigate the impact of information volume by varying the number of attributes. It would be interesting to investigate the effect of information volume on the number of choices to see whether the same results hold for preference inconsistency. Another interesting path for future research to investigate would be to reveal the psychological pathways behind our identified phenomenon linking information volume and preference inconsistency. We could use eyeball tracking experiments and cognitive processing scales to determine the proportional causes of such inconsistency: memory shortage, limited cognitive capacity, or usage of mental shortcuts such as heuristics and instincts.

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Appendix A: Attribute Clustering and Standardization

For Study 3, with the many supplier attributes included in both the short- and long-version tasks, we perform attribute clustering before running the empirical models. In Table 1, we note that our experiment participants tend to assign similar importance levels to supplier attributes that belong to the same dimension. This observation further validates our clustering choice. Because of the long list of service attributes, we further break down the service attributes into four groups based on service type, i.e., guarantee services, delivery services, replacement services, and other additional services. We list the correspondence between the service groups and the detailed service attributes in Table 8. This breakdown of service attributes is also visible to the experiment participants when they perform the supplier selection tasks. We use dashed boxes to cluster the service attributes for each candidate supplier (see Figure 3 for the display samples).

Table 1 Self-Reported Importance of Supplier Attributes

Attribute	Mean	Std. Dev.	Attribute	Mean	Std. Dev.
Price	4.268	1.094	Delivery Guarantee	2.106	1.223
Claimed Quality Level	4.617	0.789	Platform-Endorsed Delivery Guarantee	2.200	1.292
Description Accuracy	4.160	1.174	Shipping in 48 Hours	2.738	1.291
Lead Time	3.783	1.208	Replacement with No Reason in 15 Days	2.563	1.395
Response Rate	3.534	1.299	Replacement in 15 Days	2.367	1.238
Issue Rate	3.213	1.319	Replacement in 7 Days	2.366	1.273
Material Guarantee	3.469	1.291	Sell on Credit	1.909	1.254
Quality Guarantee	4.111	1.137	Free Return	2.848	1.383
Buyer Right Guarantee	2.675	1.325	Compensation If Out of Stock	2.544	1.444

Note. This table presents the mean and the standard deviation of participants' self-reported importance levels of supplier attributes. The scale of self-reported importance levels is 1 (least important) to 5 (most important).

We report in Table 1 the summary statistics of the self-reported importance level of all supplier attributes. The average self-reported importance level suggests that participants consider the price and quality dimensions (including the quality guarantee service) as the most important decision factors, followed by the speed dimension and the material guarantee service, and then the remaining service attributes.

We now discuss the standardization steps in detail. The standardization of the price factor is straightforward, but other supplier attribute dimensions are not so straightforward and thus require additional steps. The quality dimension consists of the claimed quality level and the description accuracy. These two supplier attributes are not measured in the same unit. We have three levels of claimed quality, low, medium, and high, while the description accuracy is a percentage metric. We thus first normalize the value of the two attributes and then create an average quality measure. We then perform a second standardization of the average quality measure to ensure that the quality factor included in the regression model is normalized. These two levels of standardization might raise concerns about the accuracy of the quality measure. Therefore, in the regression analysis, we present the results from 1) separately including the two quality attributes and 2) including only the combined quality factor to reassure that our findings are not determined by the standardization approach.

Table 2 Learning Effects: Robustness Checks

p-Value Threshold = 0.05	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Survey Format: LS				Survey Format: SL			
	Long - 1		Short - 2		Short - 1		Long - 2	
Panel A: Alternative Clustering of Service Factors								
No. (Pct.) of Inconsistent Rankings	16	(76%)	12	(80%)	14	(93%)	14	(67%)
No. (Pct.) of Rankings with Flipped Signs	5	(24%)	4	(27%)	1	(7%)	0	(0%)
Total No. of Pairs	21		15		15		21	

Note. We report the number of inconsistent rankings in odd columns and report the percentage of inconsistent rankings in the parentheses in even columns.

The speed dimension includes three supplier attributes: the manufacturing lead time, the response rate, and the issue rate. Because all three attributes are measured based on the percentage of satisfaction, we first take the average of the three attributes and then perform the standardization.

The approach to standardize the value of the multiple service attribute dimensions follows the same logic. Since service attributes are binary indicators that illustrate whether the candidate supplier provides the service, we first take the average of the binary indicators to obtain the percentage of services offered and then standardize the percentage metric.[2] We conduct separate standardization for the long and short versions of the supplier selection questions due to the difference in the number of available services. Our results are consistent if we conduct a uniform standardization of service attributes.

We notice that the different self-reported importance levels of the service attributes suggest that participants might form an alternative clustering of service attributes to the one we proposed based on the experiment display. To reflect on this concern, we create alternative service groups based on the self-reported importance levels. Specifically, with their high importance levels, we cluster the quality guarantee service and the material guarantee service to compose the first service group. This service group has the greatest potential impact on supplier selection outcomes. We then cluster the service attributes with average importance levels between 2.3 and 3 to compose the second service group with a medium potential impact. We cluster the remaining service attributes with lower than 2.2 average importance levels into the last service group. We present the estimation results with this alternative service group definition from the conditional logit models in Appendix Table 10. We summarize the inconsistency levels in the self-reported and the estimated actual importance levels in Panel A of Table 2. Our results are robust with this alternative clustering method.

[2]Replacement service is the only service attribute dimension that is defined differently. It is apparent that replacement with no reason in 15 days is the most valuable replacement service, followed by replacement in 15 days, and then replacement in 7 days. We thus coded the replacement service using numerical values 0–3, with 0 representing no replacement service available and 3 representing replacement with no reason in 15 days. Last, we standardize the coded replacement service.

Appendix B: Additional Tables

Table 3 Study 1: Long Version All Ten Supplier Attributes

Attribute	Choice Model Estimation (1)	Self-Report Importance Level (2)
Price	-5.507***	4.173
	(1.666)	(0.908)
Claimed Quality Level	0.643***	3.286
	(0.166)	(0.974)
Description Accuracy	0.640***	4.378
	(0.138)	(0.831)
Positive Review Rate	0.747***	4.122
	(0.211)	(0.777)
Lead Time	0.772***	3.367
	(0.253)	(0.901)
Response Rate	0.073	3.724
	(0.085)	(0.847)
Issue Rate	0.051	3.459
	(0.198)	(0.864)
Replacement in 7 Days	1.042***	3.378
	(0.136)	(1.108)
Material Guarantee	0.828***	4.204
	(0.289)	(0.941)
Logistic Warranty	0.027	3.806
	(0.095)	(1.022)
Participants	98	98
Pseudo R2	0.3307	

Note. We report the coefficient estimates from conditional logit models in column (1) and present the clustered standard errors at the participant level in parentheses. We report the average and the standard deviation of participants' reported importance rankings in columns (2). *** — 0.01 level, ** — 0.05 level, and * — 0.1 level.

Table 4 Study 1: Pairwise Comparison of Self-Reported and Estimated Importance Rankings

Importance Ranking:		(1)	(2)	(3)	(4)	(5)	(6)
		Long Version (Cluster)			Short Version		
		Choice Estimation Model		Self-Report	Choice Estimation Model		Self-Report
Attribute 1	Attribute 2	All tasks	Second Half		All tasks	Second Half	
Price	Quality	>*	>***	>**	> ***	>***	>**
Price	Speed	> ***	>***	>***	> ***	>***	>***
Price	Service	> ***	>***	>***	> ***	>***	>***
Quality	Speed	> ***	>***	>***	> ***	>***	>***
Quality	Service	> ***	>**	=	> ***	>***	>***
Speed	Service	<***	<**	<***	<***	=	>***

Note. To compare the coefficient estimates obtained from the conditional logit models, we report the p-values from chi-square tests. For self-reported rankings, we report the p-values from two-sided t-tests. *** — 0.01 level, ** — 0.05 level, and * — 0.1 level.

Table 5 Study 2: Effect of Supplier Attributes on Selection Decisions

Last 6 Long Questions	SL Version		LL Version	
Individual Attributes	Choice Model	Self-Report	Choice Model	Self-Report
	(1)	(2)	(3)	(4)
Price	-11.813***	4.156	-6.267***	4.116
	(2.826)	(0.898)	(1.226)	(0.988)
Claimed Quality Level	1.001***	3.208	0.529***	3.211
	(0.245)	(1.205)	(0.149)	(1.051)
Description Accuracy	0.705***	4.448	0.405***	4.179
	(0.127)	(0.78)	(0.091)	(0.863)
Positive Review Rate	1.568***	4.115	0.891***	4.074
	(0.292)	(0.832)	(0.186)	(0.841)
Lead Time	1.676***	3.625	0.980***	3.516
	(0.388)	(1.029)	(0.171)	(0.898)
Response Rate	0.141	3.74	0.121	3.516
	(0.09)	(0.932)	(0.078)	(0.932)
Issue Rate	0.314	3.521	0.186	3.442
	(0.287)	(0.94)	(0.175)	(0.884)
Replacement in 7 Days	1.307***	3.25	0.926***	3.379
	(0.22)	(1.105)	(0.114)	(1.044)
Material Guarantee	1.651***	4.177	0.898***	4.032
	(0.413)	(0.951)	(0.237)	(0.939)
Logistic Warranty	0.353***	3.813	0.145	3.758
	(0.091)	(0.955)	(0.096)	(0.919)
Participants	96	96	95	95
Pseudo R2	0.3246		0.2186	

Note. We report the coefficient estimates from conditional logit models in columns (1) and (3). We present the clustered standard errors at the participant level in parentheses. We report the average and the standard deviation of participants' reported importance rankings in columns (2) and (4). *** — 0.01 level, ** — 0.05 level, and * — 0.1 level.

Table 6 Study 2: Combine Price and Quality Factors

Last 6 Long Questions	SL Version		LL Version	
Individual	Choice Model	Self-Report	Choice Model	Self-Report
	(1)	(2)	(3)	(4)
Price_Quality (Combined)	1.397***	0.000	0.883***	3.897
	(0.147)	(0.550)	(0.111)	(0.497)
Lead Time	0.304***	3.625	0.333***	3.516
	(0.083)	(1.029)	(0.092)	(0.898)
Response Rate	0.073	3.74	0.151**	3.516
	(0.068)	(0.932)	(0.068)	(0.932)
Issue Rate	0.451***	3.521	0.215***	3.442
	(0.103)	(0.94)	(0.081)	(0.884)
Replacement in 7 Days	0.726***	3.25	0.694***	3.379
	(0.097)	(1.105)	(0.094)	(1.044)
Material Guarantee	0.910***	4.177	0.488***	4.032
	(0.122)	(0.951)	(0.090)	(0.939)
Logistic Warranty	0.070	3.813	-0.014	3.758
	(0.065)	(0.955)	(0.067)	(0.919)
Participants	96	96	95	95
Pseudo R2	0.2937		0.1963	

Note. We report the coefficient estimates from conditional logit models in columns (1) and (3). We present the clustered standard errors at the participant level in parentheses. We report the average and the standard deviation of participants' reported importance rankings in columns (2) and (4). *** — 0.01 level, ** — 0.05 level, and * — 0.1 level.

Table 7 Study 3: Descriptions of Supplier Attributes

Dimension	Supplier Attributes	Explanation
Price	*Price*	Unit price of the product.
Quality	*Claimed Quality Level*	A self-reported quality level of the product: high, medium, or low.
	Description Accuracy	Customer rating of the accuracy of the product description.
Speed	*Lead Time*	A platform-generated rating for the speed of product manufacturing.
	Response Rate	A platform-generated rating for measuring the supplier's speed in responding to customer requests.
	Issue Rate	A platform-generated rating for measuring the supplier's speed in shipping out finished products.
Service	*Quality Guarantee*	Whether the supplier guarantees the product quality to be consistent with the samples: if inconsistent, the supplier is required to compensate the buyer.
	Material Guarantee	Whether the supplier guarantees that the products are manufactured using the promised materials: if not, the supplier is required to compensate the buyer.
	Buyer Right Guarantee	Whether the supplier guarantees to pay up to 400-dollar (currency converted) deposit to the buyer if the supplier breaks the contract: this guarantee is endorsed by the platform.
	Replacement in 7 Days	Whether the supplier agrees to replace damaged products when the damage is not man-made: the buyer should make a request within 7 days after confirmation.
	Replacement in 15 Days	Whether the supplier agrees to replace damaged products when the damage is not man-made: the buyer should make a request within 15 days after confirmation.
	Replacement with No Reason in 15 Days	Whether the supplier allows the buyer to replace the products unconditionally within 15 days if the products are still in good condition.
	Delivery Guarantee	Whether the supplier guarantees to ship the products within the agreed time window: if not, the supplier is required to compensate the buyer.
	Platform-Endorsed Delivery Guarantee	Whether the supplier guarantees to ship the products within the agreed time window: if not, the supplier is required to compensate the buyer. This guarantee is endorsed by the platform and enables convenient compensations compared to the Delivery Guarantee.
	Shipping in 48 Hours	Whether the supplier agrees to ship the products within 48 hours after the products are manufactured, i.e., the lead time is reached.
	Sell on Credit	Whether the buyer can make a purchase on credit for at most 30 days.
	Free Return	Whether the supplier covers the logistic insurance: if yes, an insurance company will cover the logistic fees for the buyer to return the goods.
	Compensation If Out of Stock	Whether the supplier promises to compensate the buyer if the products cannot be delivered on time.

Table 8 Study 3: Supplier Attributes

Dimension	Attributes
Price	Price
Quality	Claimed Quality Level, Description Accuracy
Speed	Lead Time, Response Rate, Issue Rate
Guarantee Services	Quality Guarantee, Material Guarantee, Buyer Right Guarantee
Delivery Services	Shipping in 48 Hours, Delivery Guarantee, Platform-Endorsed Delivery Guarantee
Replacement Services	Replacement in 7 Days, Replacement in 15 Days, Replacement with No Reason in 15 Days
Additional Services	Sell on Credit, Free Return, Compensation If Out of Stock

Note. The six service attributes marked in blue are not present in the short version.

Table 9 Study 3: Pairwise Comparison of Self-Reported and Estimated Importance Rankings

Importance Ranking:		(1)	(2)	(3)	(4)	(5)	(6)
			Long Version			Short Version	
Attribute 1	Attribute 2	Self-Report	Both Sets	First Set	Self-Report	Both Sets	First Set
Price	Quality	=	<***	<***	=	<***	<*
Price	Speed	>***	=	=	>***	=	=
Price	Guarantee Service	>***	<***	<***	>***	<***	=
Price	Delivery Service	>***	=	<***	>***	<***	=
Price	Replacement	>***	=	<**			
Price	Additional Service	>***	<**	<**	>***	<***	=
Quality	Speed	>***	>***	>*	>***	>**	=
Quality	Guarantee Service	>***	=	<**	>***	=	=
Quality	Delivery Service	>***	>***	=	>***	=	=
Quality	Replacement	>***	>***	=			
Quality	Additional Service	>***	>**	=	>***	=	=
Speed	Guarantee Service	=	<***	<***	=	<***	=
Speed	Delivery Service	>***	<**	<***	>***	<**	=
Speed	Replacement	>***	<*	<**	>***		
Speed	Additional Service	>***	<***	=	>***	<***	=
Guarantee Service	Delivery Service	>***	>***	>***	>***	=	=
Guarantee Service	Replacement	>***	>***	>***			
Guarantee Service	Additional Service	>***	>***	>***	>***	=	>**
Delivery Service	Replacement	=	=	=			
Delivery Service	Additional Service	=	=	=	>**	=	=
Replacement	Additional Service	=	=	=			

Note. For self-reported rankings, we report the p-values from two-sided t-tests. For the comparison of the coefficients estimates obtained from the conditional logit models, we report the p-values from chi-square tests. *** — 0.01 level, ** — 0.05 level, and * — 0.1 level.

Table 10 Study 3: Effect of Learning in Selection Decisions: Alternative Clustering of Factors

Dependent Variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Selection Outcome	Alternative Cluster of Service Factors				Combined Price and Quality Factors			
	Survey Format: LS		Survey Format: SL		Survey Format: LS		Survey Format: SL	
	Long - 1	Short - 2	Short - 1	Long - 2	Long - 1	Short - 2	Short - 1	Long - 2
Price	-0.192	-0.263*	-0.264**	-0.453***				
	(0.125)	(0.150)	(0.126)	(0.125)				
Quality	0.757***	0.796***	0.492***	0.661***				
	(0.202)	(0.245)	(0.116)	(0.134)				
Price_Quality (combined)					0.265***	0.200**	0.358***	0.572***
					(0.083)	(0.092)	(0.076)	(0.088)
Speed	-0.005	-0.040	0.351***	0.358***	0.078	0.002	0.438***	0.345***
	(0.066)	(0.061)	(0.062)	(0.080)	(0.090)	(0.054)	(0.071)	(0.080)
Service 1/	0.539***	0.301***	0.579***	0.536***	0.634***	0.159***	0.511***	0.743***
Guarantee Service	(0.098)	(0.074)	(0.075)	(0.088)	(0.168)	(0.051)	(0.062)	(0.120)
Service 2/	0.495***	0.602***	0.403***	0.261***	0.431***	0.417***	0.484***	0.174
Delivery Service	(0.070)	(0.085)	(0.087)	(0.082)	(0.082)	(0.058)	(0.086)	(0.123)
Replacement Service	0.410***			0.228**	0.320***			0.205*
	(0.105)			(0.093)	(0.071)			(0.112)
Service 3/	0.201***	0.206**	0.392***	0.302***	0.423***	0.438***	0.334***	0.454***
Additional Service	(0.074)	(0.095)	(0.046)	(0.068)	(0.080)	(0.061)	(0.056)	(0.074)
Participants	80	80	82	82	80	80	82	82
Observations	1,440	1,440	1,476	1,476	1,440	1,440	1,476	1,476
Pseudo R2	0.179	0.169	0.205	0.267	0.160	0.153	0.210	0.264

† *Note.* *** — 0.01 level, ** — 0.05 level, and * — 0.1 level. In the first four columns, we report the results from using an alternative cluster of service factors. In the last four columns, we report the results after combining the price and quality factors. We present the clustered standard errors at the participant level in the parentheses.

Table 11 Study 3: Effect of Learning in Selection Decisions: Satisficers and Maximizers

Dependent Variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Selection Outcome	Satisficers				Maximizers			
	Survey Format: LS		Survey Format: SL		Survey Format: LS		Survey Format: SL	
	Long - 1	Short - 2	Short - 1	Long - 2	Long - 1	Short - 2	Short - 1	Long - 2
Price	0.015	-0.062	-0.399**	-0.377**	-0.087	-0.081	-0.343*	-0.732***
	(0.199)	(0.202)	(0.171)	(0.162)	(0.141)	(0.153)	(0.201)	(0.213)
Quality	0.494*	0.574**	0.628***	0.891***	0.548***	0.236	0.422**	0.955***
	(0.292)	(0.254)	(0.143)	(0.200)	(0.192)	(0.209)	(0.189)	(0.223)
Speed	0.081	0.074	0.398***	0.286***	0.300***	-0.016	0.485***	0.346***
	(0.155)	(0.089)	(0.106)	(0.111)	(0.109)	(0.079)	(0.116)	(0.132)
Guarantee Service	0.627**	0.173**	0.437***	0.500***	1.302***	0.210***	0.605***	1.018***
	(0.288)	(0.076)	(0.094)	(0.168)	(0.209)	(0.070)	(0.088)	(0.189)
Delivery Service	0.331***	0.340***	0.484***	0.298**	0.554***	0.388***	0.494***	-0.121
	(0.121)	(0.100)	(0.132)	(0.144)	(0.099)	(0.070)	(0.131)	(0.210)
Replacement Service	0.369***			0.363**	0.398***			0.055
	(0.120)			(0.162)	(0.105)			(0.146)
Additional Service	0.428***	0.440***	0.339***	0.504***	0.314***	0.440***	0.329***	0.517***
	(0.126)	(0.080)	(0.075)	(0.126)	(0.100)	(0.085)	(0.093)	(0.113)
Participants	34	34	41	41	46	46	41	41
Observations	612	612	738	738	828	828	738	738
Pseudo R2	0.171	0.184	0.191	0.234	0.200	0.157	0.236	0.323

Note. In the first four columns, we report the results from satisficer participants. In the last four columns, we report the results from maximizer participants. We present the clustered standard errors at the participant level in the parentheses. *** — 0.01 level, ** — 0.05 level, and * — 0.1 level.